

Supplementary material: Support-Aware Review Queues for Crop–Weed Vision: A Multitemporal Single-Field Case Study

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S1. Public-data lineage and official spatial protocol

This study uses two public datasets. SugarBeets2016 supplies source-specialist supervision, and WeedsGalore supplies target development and evaluation data. CropAndWeed and PhenoBench formed preliminary data-lineage audits and remain outside the reported experiments.

Table 1: Official WeedsGalore split by acquisition date. Instance counts are from released raster enumeration.

Date	Train tiles	Validation tiles	Test tiles	Test crop	Test weed
25 May 2023	32	8	8	122	392
30 May 2023	32	8	8	55	440
6 June 2023	32	8	8	68	726
15 June 2023	8	2	2	11	154
Total	104	26	26	256	1,712

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The public paper describes one field, 48 recorded locations, and an 8/2/2 spatial-patch allocation. The release supplies official split labels and capture indices. The manifest preserves this released spatial structure. Across all 156 rasters, enumeration yields 2,168 crop and 10,029 weed instances; the dataset paper reports 2,169 and 10,031. The consistency audit found zero role disagreements between instance and semantic rasters and zero excluded identifiers. Archive and manifest SHA-256 checksums are recorded in the reproducibility package.

S2. Annotation and information-flow ledger

Table 2: Supervision ledger. “Available” distinguishes labels present in the public data from labels actually read by a stage.

Information	Available	Read by	Purpose
SugarBeets pixel masks	Public source	Specialist training	Train foreground and semantic specialists
Target train dense masks	104 tiles	Proposal audit; label derivation	Select the commissioned proposal setting and derive candidate roles
Target validation dense masks	26 tiles	Model/queue selection	Select C , fine-tuning epoch, semantic epoch, threshold or area, or queue-score threshold
Target test dense masks	26 tiles	Offline scorer after declared outputs	Compute proposal, queue, IoU, and crop-overlap metrics
300 mask-derived training-candidate labels/draw	Simulated reveal from train pool	Label-budget ranker	Compare prospective acquisition policies
Proposal-generation inputs	RGB and source model outputs	Proposal generator	Create connected-component candidates

The full-supervision linear rankers use 9,887 official-training and 2,436 validation candidate labels before refitting. The equal-target-mask-access semantic baseline trains on all 104 target-training dense masks. The nominal 300-label experiment restricts only the sampled training candidates; all 2,436 labelled validation candidates remain available for model selection. Candidate labels are deterministically derived from dense masks: candidates satisfying the role-qualified weed contract are positive and all remaining candidates are negative. Prospective uniform and DINOv2 coreset strategies select training IDs before revealing labels. Retrospective class stratification uses labels during selection and serves as a retrospective reference.

The core configuration comprises the split, proposal grid, matching contract, feature definitions, and discrimination-selected rankers archived before final official-test scoring. Target-semantic, repeated-seed, quota, exact-mask, per-tile, crop-severity, and date/patch analyses were specified after the core test had been accessed and are exploratory. Training and validation data selected regularization, thresholds, epochs, and candidate geometry. Each test policy formed its queue from images, scores, tile identities, and declared group labels before the offline scorer accessed test masks.

Source specialist training and source-domain diagnostics

Both source specialists used a ResNet-18 U-Net initialized from torchvision ImageNet-1K V1 weights, ImageNet RGB normalization, AdamW, bfloat16 autocasting, gradient clipping at 5, and the 7,050-image SugarBeets source training partition. The foreground specialist used 32,768 deterministic 128×128 patches, BCE with positive weight 30 plus soft Dice, and 10 epochs. The three-class semantic specialist used 16,384 role-balanced patches, weighted cross-entropy (0.2, 1, 1) plus plant-role Dice, and 8 epochs. Training used deterministic patches, and the final epoch supplied each checkpoint. The 901-image source validation partition subsequently provided operating-point diagnostics.

On the later source validation diagnostic, the foreground model’s maximum scanned pixel IoU was 0.3821 at threshold 0.99 (precision 0.5788, recall 0.5293); threshold 0.70 gave precision 0.0479 and recall 0.9631. The semantic model had raw three-class macro IoU 0.4425. Its independently selected role threshold produced balanced accuracy 0.7224, crop recall 0.5524, and weed recall 0.8924. These metrics document source behavior and checkpoint provenance; target queue conclusions use the official WeedsGalore split.

S3. Proposal selection, definitions, and burden

For truth instance I , candidate C , labeled pixels L , and pixels of role R , the audit records

$$q_I = |I \cap C|/|I|, \quad q_L = |C \cap L|/|C|, \quad q_R = |C \cap R|/|C \cap L|.$$

Spatial recall requires $q_I \geq 0.50$. Role-qualified recall additionally requires $q_L \geq 0.50$, $q_R \geq 0.90$, and a role match. Queued recall restricts candidates to the selected queue. These fixed engineering thresholds define offline candidate accounting rather than agronomic treatment or actuation; the evaluated 27-cell sensitivity grid varies all three values.

Table 3: Official-training proposal grid. Feasible settings have training-set mean burden at most 100 candidates/image, with tile-level counts unrestricted.

Threshold / minimum area (px)	Candidates/image	Spatial weed recall	Qualified weed recall	Feasible
0.30 / 8	281.35	0.7135	0.2406	No
0.30 / 32	142.04	0.7000	0.2297	No
0.50 / 8	259.49	0.6325	0.2822	No
0.50 / 32	123.52	0.6190	0.2707	No
0.70 / 8	207.48	0.5416	0.2979	No
0.70 / 32	95.07	0.5271	0.2856	Yes
0.80 / 8	168.63	0.4806	0.2959	No
0.80 / 32	76.99	0.4672	0.2835	Yes
0.90 / 8	115.50	0.3703	0.2421	No
0.90 / 32	52.38	0.3587	0.2317	Yes
0.95 / 8	75.71	0.2452	0.1685	Yes
0.95 / 32	36.83	0.2366	0.1605	Yes

The threshold 0.70, 32-pixel setting maximized training qualified recall among feasible settings. The deterministic tie order was qualified recall, lower mean burden, higher threshold, then larger minimum area. The earlier externally fixed setting used threshold 0.95 and the same minimum area and provides the shared-ranker-weight comparison.

Candidate burden under the original setting had median 28.5, interquartile range 19.25–43.75, maximum 62, and 7/26 test images with fewer than 20 candidates. The commissioned setting had median 89, interquartile range 60–114.25, maximum 183, 10/26 images above 100, and no image with fewer than 20 candidates. Its 30 May mean was 130.38 on training and 119.38 on test. Candidate composition changed from 453 weed/148 crop/201 ambiguous (802 total) to 464/111/1,776 (2,351 total).

Table 4: Official-test proposal layers by date. Numerators are weed instances.

Date	Setting	Weed n	Spatial recalled	Qualified recalled	Candidates/image
25 May	Original	392	2 (0.0051)	2 (0.0051)	13.88
25 May	Commissioned	392	51 (0.1301)	13 (0.0332)	102.75
30 May	Original	440	106 (0.2409)	84 (0.1909)	35.75
30 May	Commissioned	440	263 (0.5977)	121 (0.2750)	119.38
6 June	Original	726	328 (0.4518)	237 (0.3264)	45.13
6 June	Commissioned	726	566 (0.7796)	369 (0.5083)	64.63
15 June	Original	154	73 (0.4740)	7 (0.0455)	22.00
15 June	Commissioned	154	98 (0.6364)	15 (0.0974)	28.50
Pooled	Original	1,712	509 (0.2973)	330 (0.1928)	30.85
Pooled	Commissioned	1,712	978 (0.5713)	518 (0.3026)	90.42

Table 5: Split/merge sensitivity across the complete 156-tile release. “Merge” counts candidates overlapping at least two instances at the stated instance-coverage threshold.

Setting	Threshold	Split instances	Merge candidates
Original	0.10	224	735
Original	0.25	40	640
Original	0.50	0	473
Commissioned	0.10	154	1,136
Commissioned	0.25	26	1,065
Commissioned	0.50	0	938

S4. Fixed-weight proposal comparison

One all-candidate role-plus-local logistic ranker was fitted on commissioned train plus validation candidates. Its scaler and coefficients were then applied unchanged to both the original and commissioned test proposals. This isolates proposal-pool changes from ranker refitting.

Table 6: Shared-ranker-weight comparison at $K = 20$. Ranker weights are identical, but workloads differ because seven original-setting tiles supply fewer than 20 candidates. Crop frequency uses all 26 test tiles as denominator.

Setting	Accepted	Precision	Spatial	Qualified	1:1 qualified	Any crop
Original	471	276/471	321/1712	205/1712	166/1712	22/26
Commissioned	520	247/520	467/1712	273/1712	201/1712	23/26

The complete $K = 1, 5, 10, 20, 50$ curves and per-date rows are supplied in the reproducibility archive. All ties are resolved by lexicographically ascending

candidate ID.

S5. All-candidate ranking, per-date queue results, and ceilings

Table 7: Candidate-ranking support audit on the 2,351 official-test candidates.

Model and fitting support	AUC	AP
Source ResNet score, all candidates	0.4947	0.2877
Eligible role+local, all-candidate test	0.4420	0.1640
All-candidate role+local	0.9049	0.6960
All-candidate geometry	0.8154	0.5418
Eligible frozen DINOv2, all-candidate test	0.6321	0.3816
All-candidate frozen DINOv2	0.9630	0.8684
All-candidate masked DINOv2	0.9665	0.8813

Table 8: Component-masked DINOv2 queue at $K = 20$ by date.

Date	Accepted	Weed cand.	Precision	Spatial recall	Qualified recall	Any crop
25 May	160	13	0.0813	32/392 (0.0816)	12/392 (0.0306)	8/8
30 May	160	89	0.5563	128/440 (0.2909)	84/440 (0.1909)	2/8
6 June	160	150	0.9375	220/726 (0.3030)	210/726 (0.2893)	0/8
15 June	40	26	0.6500	79/154 (0.5130)	13/154 (0.0844)	1/2
Pooled	520	278	0.5346	459/1712 (0.2681)	319/1712 (0.1863)	11/26

For frozen DINOv2 at $K = 20$, the role-qualified proposal ceiling is 518/1,712, the exact unconstrained oracle is 426/1,712, and the discrimination-selected masked queue is 319/1,712. The corresponding ranking gap is 0.0625. For a maximum crop-overlap scene frequency of 0.01, $\lfloor 0.01 \times 26 \rfloor = 0$ scenes are allowed. Crop-excluding optimization yields 418/1,712 recalled instances. Every ceiling uses exact integer optimization. The 1,000 archived random-ranking draws have $K = 20$ 2.5/97.5 percentiles of 0.077/0.104.

Validation-queue selection and allocation-policy analysis

The queue analysis selected logistic regularization by validation $K = 20$ one-to-one qualified recall, followed by many-to-one recall, precision, and C closest to 1 on the log scale. The selected value was $C = 10$. The model was refitted on train plus validation before test scoring. The post-test 5–50 policy maximized total score under a 520-candidate budget and is reported as exploratory. Unconstrained global top-520 is a diagnostic upper bound. The

validation threshold is the boundary score of the validation global top-520 queue; transferring that fixed score to test accepted 426 candidates.

Candidate-level score discrimination was heterogeneous before allocation. Pooled AUC/AP were 0.9605/0.8617, whereas date-macro values were 0.8774/0.6901. Table 9 reports each date. Twenty-four of 26 tiles contained both classes and supported tile-level AUC/AP; their macro values were 0.9091/0.6869, with worst-tile values of 0.6667/0.0263. These values are descriptive and are not treated as independent population replicates.

Table 9: Candidate-level role-qualified discrimination for the queue-selected ranker by acquisition date. Class counts accompany each AUC/AP value.

Stratum	Candidates	Weed positive	Other	AUC	Average precision
25 May	822	15	807	0.8798	0.2131
30 May	955	132	823	0.9535	0.7492
6 June	517	279	238	0.9381	0.9402
15 June	57	38	19	0.7382	0.8577
Pooled	2,351	464	1,887	0.9605	0.8617
Date macro	—	—	—	0.8774	0.6901

Table 10: Queue allocation using the validation-queue-selected masked-DINOv2 ranker. The post-test 5–50/tile and unconstrained global top- N policies are exploratory diagnostics.

Policy	Accepted	Precision	Spatial	Qualified	1:1 qualified	Any crop
Fixed $K = 20$	520	275/520	390/1712	324/1712	227/1712	13/26
Exploratory 5–50/tile	520	382/520	533/1712	453/1712	315/1712	14/26
Diagnostic global top- N	520	388/520	547/1712	464/1712	320/1712	12/26
Validation threshold	426	348/426	485/1712	424/1712	290/1712	8/26

Table 11: Per-date one-to-one qualified recall and accepted candidates for fixed, exploratory 5–50, and diagnostic global allocation.

Date	Fixed N	Fixed recall	Bounded N	Bounded recall	Global N	Global recall
25 May	160	10/392 (0.0255)	51	6/392 (0.0153)	38	6/392 (0.0153)
30 May	160	71/440 (0.1614)	149	77/440 (0.1750)	153	78/440 (0.1773)
6 June	160	135/726 (0.1860)	285	221/726 (0.3044)	293	225/726 (0.3099)
15 June	40	11/154 (0.0714)	35	11/154 (0.0714)	36	11/154 (0.0714)

For global minus fixed one-to-one qualified recall, the 26 paired tile differences had median 0, interquartile range 0–0.0763, and range -0.1176 to

0.2027. Nine tiles improved, 12 were unchanged, and 5 declined. Many-to-one differences had median 0, interquartile range 0–0.1000, and range -0.1176 to 0.3919, with the same 9/12/5 counts. Two capture-index proxy groups yielded global qualified recall of 0.2574 and 0.2806; corresponding fixed values were 0.1782 and 0.1970.

Table 12: Allocation concentration across 26 test tiles. The 6 June share is the fraction of each policy’s accepted candidates assigned to that acquisition date.

Policy	Min/tile	Zero tiles	Tiles < 5	Tiles < 10	Max/tile
Fixed $K = 20$	20	0	0	0	20
Exploratory 5–50	5	0	0	8	50
Diagnostic global	0	1	5	8	55
Validation threshold	0	2	7	11	45

Policy	Allocation Gini	6 June share
Fixed $K = 20$	0.0000	0.3077
Exploratory 5–50	0.3817	0.5481
Diagnostic global	0.4220	0.5635
Validation threshold	0.4648	0.5986

Table 13: Date-macro and worst-date queue metrics. Worst denotes the minimum across the four dates.

Policy	Macro P	Macro Q	Macro 1:1	Worst Q	Worst 1:1
Fixed $K = 20$	0.5516	0.1504	0.1111	0.0255	0.0255
Exploratory 5–50	0.6133	0.1959	0.1415	0.0153	0.0153
Diagnostic global	0.6160	0.2001	0.1435	0.0153	0.0153
Validation threshold	0.6876	0.1836	0.1311	0.0128	0.0128

Table 14: Crop exposure severity. Crop-instance rows count 256 official-test crop instances.

Policy	Crop-overlap cand.	Crop frac. 95th	Crop frac. max	Crop inst. ≥ 0.10	≥ 0.25	≥ 0.50
Fixed $K = 20$	57/520	0.5910	0.9730	52/256	50/256	32/256
Diagnostic global top- N	26/520	0.0036	0.9583	21/256	19/256	10/256
Exploratory 5–50	28/520	0.1148	0.9189	23/256	20/256	12/256
Validation threshold	14/426	0	0.9189	12/256	10/256	5/256

Equal-target-mask-access semantic baseline

The archived 40-epoch distributed reference run followed its specified split, optimizer, augmentation, threshold grid, area grid, and selection rule. Validation selected epoch 36, threshold 0.30, and 16-pixel minimum area. Validation three-class/plant-role mean IoU were 0.7637/0.6582; fixed- K one-to-one qualified recall was 0.1908. Official-test semantic IoU was 0.9778 background, 0.6957 crop, and 0.7131 weed (three-class mean 0.7955; plant-role mean 0.7044). A separate five-seed H200 series quantifies training variability under the same single-device recipe.

Table 15: Official-test target-semantic queue. Components are ranked by their target-semantic weed score; validation alone selected epoch and component settings.

Model	Candidates	Accepted	Weed cand.	Precision	Spatial	Qualified	1:1 qualified	Crop scenes
Target semantic, fixed $K = 20$	1,347	518	355	0.6853	520/1712 (0.3037)	453/1712 (0.2646)	348/1712 (0.2033)	6/26 (0.2308)

Table 16: Complete representation-allocation comparison. Exploratory 5–50 policies preserve the fixed total within each representation.

Representation	Allocation	N	Prec.	1:1	Crop	Worst 1:1
Masked DINO	Fixed 20/tile	520	0.5288	0.1326	13/26	0.0255
Masked DINO	Exploratory 5–50	520	0.7346	0.1840	14/26	0.0153
Target semantic	Fixed 20/tile	518	0.6853	0.2033	6/26	0.1169
Target semantic	Exploratory 5–50	518	0.8205	0.2477	1/26	0.0510

Five-seed H200 stability series

Five independently initialized single-device runs used the same 40-epoch recipe, official split, validation-only selection rule, and batch size of six. All runs completed on NVIDIA H200 devices. Validation selected the 16-pixel component minimum in every run; the foreground threshold varied across 0.30, 0.50, and 0.70. Table 17 reports every seed rather than treating candidates, tiles, or plant instances as independent training repetitions.

The fixed queue accepted 511.8 ± 5.6 candidates across seeds (mean $\pm$ sample SD). Fixed precision and one-to-one recall were 0.7168 ± 0.0260 and 0.2027 ± 0.0057 ; exploratory 5–50 allocation increased them to 0.8939 ± 0.0249 and 0.2578 ± 0.0090 . The paired 5–50-minus-fixed differences were $+0.1770 \pm$

Table 17: Per-seed target-semantic H200 results. Seed values are numeric identifiers, not acquisition dates. Fixed precision and one-to-one recall use $K = 20$ per tile. The exploratory 5–50 allocation preserves the fixed accepted total within each seed.

Seed ID	Epoch	T	A	Semantic mIoU	Fixed prec.	Fixed 1:1	5–50 prec.	5–50 1:1
20260810	38	0.5	16	0.7931	0.7112	0.2062	0.8841	0.2593
20260811	36	0.7	16	0.7790	0.7248	0.1992	0.9188	0.2605
20260812	40	0.3	16	0.8116	0.6879	0.2033	0.8786	0.2605
20260813	39	0.5	16	0.7897	0.7569	0.2097	0.9216	0.2664
20260814	31	0.5	16	0.8041	0.7035	0.1951	0.8663	0.2424

0.0146 for precision and $+0.0551 \pm 0.0053$ for one-to-one recall, and both were positive in all five seeds. Crop-exposed scene frequency decreased by 0.1846 ± 0.0632 in all five seeds. Worst-date one-to-one recall decreased by 0.0627 ± 0.0070 in all five seeds.

For the stability analysis, each selected checkpoint was replayed on the official test images to regenerate component masks and candidate–instance edges. Candidate IDs and fixed-queue precision, many-to-one recall, one-to-one recall, and crop exposure were checked against the archived run output for every seed and every test tile before alternative allocation policies were evaluated. The reference replay regenerated all 1,347 candidate identifiers and reproduced all 26 fixed-queue scene records. The archive also preserves an unsuccessful box-only reconstruction that overcounted one recovered instance in one seed.

Table 18: Complete post-test quota grid over the five H200 seeds. Each cell reports mean pooled / mean worst-date one-to-one role-qualified recall. “None” denotes no maximum.

Minimum	Maximum 20	Maximum 30	Maximum 50	Maximum none
0	0.2027 / 0.1100	0.2484 / 0.0558	0.2643 / 0.0243	0.2643 / 0.0210
5	0.2027 / 0.1100	0.2468 / 0.0583	0.2578 / 0.0473	0.2578 / 0.0473
10	0.2027 / 0.1100	0.2405 / 0.0809	0.2442 / 0.0783	0.2439 / 0.0783
20	0.2027 / 0.1100	0.2027 / 0.1100	0.2027 / 0.1100	0.2027 / 0.1100

Validation policy selection required validation candidate-level role-qualified precision at least as high as fixed $K = 20$, then maximized validation worst-date one-to-one recall, pooled one-to-one recall, and precision. It selected 0–20 in seed IDs 20260810, 20260812, 20260813, and 20260814, and 0–30 in seed ID 20260811. Thus, the exploratory 5–50 setting was not selected by the

group-aware validation rule. Date-balanced global allocation yielded precision 0.7710 ± 0.0226 , pooled one-to-one recall 0.2192 ± 0.0064 , and worst-date recall 0.1163 ± 0.0190 . Its paired one-to-one effect relative to fixed allocation was negative in all five seeds for capture-index patch proxy 0 (mean -0.0458) and positive in all five for proxy 1 (mean $+0.0603$). The 5–50 effect was positive in both proxies for all five seeds.

IoU-only weed-instance recall at thresholds 0.25/0.50/0.75 was 429/1,712 (0.2506), 280/1,712 (0.1636), and 53/1,712 (0.0310). The median, 75th percentile, and 95th percentile queued crop fraction were all zero; the maximum was 0.9474. Two tiles contained fewer than 20 components, so the accepted total was 518 rather than 520.

Table 19: Target semantic fixed- K queue by acquisition date. Every proportion is accompanied by an archived per-tile integer count.

Date	Tiles	Candidates	Accepted	Precision	Spatial	Qualified	1:1 qualified	Crop scenes
25 May	8	304	158	70/158 (0.4430)	88/392 (0.2245)	68/392 (0.1735)	63/392 (0.1607)	6/8
30 May	8	446	160	124/160 (0.7750)	175/440 (0.3977)	144/440 (0.3273)	124/440 (0.2818)	0/8
6 June	8	534	160	143/160 (0.8938)	226/726 (0.3113)	211/726 (0.2906)	143/726 (0.1970)	0/8
15 June	2	63	40	18/40 (0.4500)	31/154 (0.2013)	30/154 (0.1948)	18/154 (0.1169)	0/2

Across the four dates, target-semantic macro precision, many-to-one spatial recall, many-to-one qualified recall, and one-to-one qualified recall were 0.6404, 0.2837, 0.2465, and 0.1891. The corresponding worst-date values were 0.4430, 0.2013, 0.1735, and 0.1169. Crop-overlap scene frequency had a date macro of 0.1875 and a worst-date value of 0.7500. The repeated H200 series evaluates the same metric family across independent initializations.

The date pattern differs from the commissioned masked-DINO queue. Target semantic training improves the 25 and 30 May fixed queues, while the two-image 15 June stratum reaches 0.1169 one-to-one recall. These date strata describe temporal variation within the study field.

S6. Matching-contract, IoU, crop-exposure, and multiplicity sensitivity

The evaluated queue-contract table contains every combination of model, $K \in \{1, 5, 10, 20, 50\}$, date/pooled stratum, instance coverage $\{0.25, 0.50, 0.75\}$, labeled coverage $\{0.25, 0.50, 0.75\}$, and purity $\{0.50, 0.75, 0.90\}$. For component-masked DINOv2 at $K = 20$, pooled many-to-one qualified recall ranges 0.0485–0.2886, and one-to-one recall

ranges 0.0374–0.1881. At the main 0.50/0.50/0.90 contract, these values are 319/1,712 (0.1863) and 230/1,712 (0.1343). Under IoU-only matching at thresholds 0.25/0.50/0.75, many-to-one recall is 325/1,712, 163/1,712, and 24/1,712; one-to-one recall is 307/1,712, 163/1,712, and 24/1,712.

Table 20: Crop-overlap scene frequency at $K = 20$. A scene is counted when at least one queued region has crop fraction greater than or equal to the listed threshold; for threshold zero, positive overlap is required in implementation.

Model	Any > 0	≥ 0.01	≥ 0.10	≥ 0.50
Frozen DINOv2	10/26	10/26	10/26	9/26
Masked DINOv2	11/26	11/26	11/26	9/26

The archive additionally reports queued candidate–instance multiplicity and IoU matching at 0.25, 0.50, and 0.75. These tables quantify candidates covering several small weeds and weeds fragmented across several candidates, complementing the main many-to-one metric.

S7. Restricted training-label draws with fully labelled validation and fine-tuning

Table 21: Mean official-test results over ten draws of 300 simulated training-candidate labels derived from dense masks. All models also use the fully labelled validation candidate set.

Selection	AUC	AP	$K = 20$ precision	$K = 20$ qualified recall
Prospective uniform	0.9549	0.8544	0.5242	0.1724
Prospective DINOv2 coreset	0.9543	0.8534	0.5217	0.1717
Retrospective stratified	0.9547	0.8502	0.5231	0.1725

All ten candidate-ID lists and their SHA-256 checksums are supplied in the reproducibility archive. Coreset selection is greedy farthest-first in frozen DINOv2 embedding space with a seeded initial point; uniform selection is seeded sampling without replacement. Training data determine candidate IDs, and official validation selects C and epoch. The 2.5/97.5 draw percentiles reported below are descriptive empirical quantiles, not confidence intervals.

Table 22: Paired last-block fine-tuning differences (fine-tuned minus frozen) on identical coreset draws.

Metric	Mean difference	Empirical 2.5th	Empirical 97.5th
AUC	−0.0396	−0.0525	−0.0170
$K = 20$ precision	−0.0373	−0.0476	−0.0169
$K = 20$ qualified recall	−0.0115	−0.0234	−0.0001

Table 23: Per-repeat paired AUC differences for last-block fine-tuning minus the frozen representation. Repeats use identical selected training IDs within each pair.

Repeat	0	1	2	3	4	5	6	7	8	9
Δ AUC	−0.0508	−0.0367	−0.0412	−0.0530	−0.0424	−0.0456	−0.0116	−0.0353	−0.0400	−0.0392

The predefined fine-tuning configuration used DINOv2 ViT-S/14, tight unpadded crops, 8 epochs, batch 64, AdamW, and class-weighted binary cross-entropy. Weight decay was 10^{-4} , the last-block/norm learning rate was 10^{-5} , and the head learning rate was 10^{-3} . Horizontal and vertical flips supplied augmentation. The final block, final normalization, and head were trainable, while earlier blocks remained frozen. Validation selected epoch 8 in every repeat. The paired estimates characterize this archived recipe.

S8. Statistical interpretation

The main tables report raw test numerators and denominators, pooled values, all four acquisition dates, and paired tile differences. The 15 June date contains two official-test images and consequently receives greater relative weight in an unweighted date macro. The archive supplies both date-macro and tile/instance-weighted summaries. Global-versus-fixed comparisons include the paired median, interquartile range, full range, and improved/unchanged/worsened counts. These statistics describe the official spatial split.

The release supplies capture indices as its spatial coordinate proxy. A full-tile DINOv2 Hungarian linkage to nominal locations produced negative assignment margins for 12 of 18 test assignments. We therefore used the largest ordered-index gap to form two 13-image capture-index proxy groups. Their bootstrap summaries quantify sensitivity to this grouping. Repeat-wise 2.5th and 97.5th percentiles are empirical simulation quantiles.

S9. Development diagnostics

Table 24: Date-grouped development diagnostics. The official spatial analysis is reported separately.

Analysis	Result	Interpretation
Nested Platt calibration	Date-macro Brier 0.2112 before, 0.2776 after; NLL 0.6338 before, 0.8239 after	Date transfer increased both calibration errors; queue analyses use ranking scores
Empirical threshold controller	Accepted precision 0.1317, empty response 0.7448, crop-scene frequency 0.1042	Training-date threshold transfer produced a sparse held-out queue with low precision
Cost-sensitive ranker, $K = 20$	AUC 0.7301, AP 0.7272, precision 0.4327, instance recall 0.0842, crop-scene frequency 0.3542	Crop weighting shifted recovery and exposure in opposite directions
Role $\times$ support, $K = 20$	AUC 0.7151 versus 0.8542 role-only; recall 0.0910 versus 0.0520; crop frequency 0.3698 versus 0.1510	Support multiplication raised recovery and crop exposure while weakening ranking
Last-block DI-NOv2 fine-tuning	Mean paired AUC -0.0396 and qualified recall -0.0115	Small target supervision favored frozen representations

The run registry retains every diagnostic, including unsuccessful executions, under its original identifier. Crop-overlap frequency is defined consistently across these records.

S10. Runtime, environment, and model provenance

Table 25: Principal runtime and memory records.

Component	Wall time	Peak CUDA memory	Device
Unmasked DINOv2 embeddings	93.94 s	601.9 MB	H200, physical GPU 0
Masked DINOv2 embeddings	32.52 s	601.9 MB	H200, physical GPU 0
Official proposal audit	35.46 s	CPU	–
Queue-policy and tile-level audit	7.63 s	CPU	–
Target semantic training, validation, and finalization	2.86 h	See run history	H200, physical GPUs 5–7
Target semantic test inference	0.931 s (0.0358 s/tile)	583.8 MB	H200, physical GPU 5
Five-seed target-semantic training	712.48 $\pm$ 8.22 s/run	See run history	H200, one device/run on physical GPUs 6–7
Five-seed target-semantic test replay	0.950 $\pm$ 0.021 s/run	513.4 MiB/run	H200, one device/run on physical GPUs 6–7
Ten fine-tuning repeats	93.89 s elapsed	660.1 MB/repeat	H200, physical GPUs 0–7

Table 26: Supervision and computation for the two representation paths. Runtime entries are measured records rather than hardware-normalized estimates.

Path	Target-mask access	Additional labelled data	Public pretraining	Recorded computation
Source specialists + masked DINO	104 train masks for proposal selection and 9,887 candidate labels; 26 validation masks and 2,436 candidate labels for selection	7,050 SugarBeets training masks; 901 source-validation masks	ImageNet-1K ResNet-18; DINOv2 ViT-S/14 LVD-142M	Masked test embeddings: 32.52 s, 601.9 MB; source training time was not retained
Target semantic	104 train masks; 26 validation masks for epoch and component selection	None	ImageNet-1K ResNet-18	Training, validation, and finalization: 2.86 h on three H200s; test: 0.931 s, 583.8 MB
Target-semantic stability series	Same 104 train and 26 validation masks in every seed	None	ImageNet-1K ResNet-18	Five single-device 40-epoch runs: 712.48 ± 8.22 s/run; exact test replay: 0.950 ± 0.021 s/run and 513.4 MiB

The 2.86-h distributed reference and the approximately 712-s single-device repetitions are records from different implementations and are not a hardware speed benchmark. The reference used a three-process distributed path with local batch size two and performed full validation setting search and finalization during its 40-epoch run. The later repetitions used a single-process path with local batch size six and were launched concurrently across two H200 accelerators. Model architecture, global batch size, optimization objective, epoch count, and validation selection contract were held constant; process topology, validation orchestration, and logging overhead were not. The archive therefore reports both measurements without converting them into a speedup claim.

The recorded GPU environment is Python 3.10.12, PyTorch 2.7.1+cu126, timm 1.0.28, and NumPy 2.2.6. The public DINOv2 ViT-S/14 LVD-142M weights were obtained through timm. Model, embedding, configuration, and output checksums are recorded in the archive manifest.

S11. Reproducibility and archive record

The versioned archive is available at <https://github.com/Fytap/argrispec-vlm-reproducibility/releases/tag/v1.1.0>. It preserves

the executable directory structure, frozen configurations, declared inputs, source-data tables, environment specification, run manifests, and unsuccessful execution records. Clean-room commands recompute tables and figures from archived predictions and exact matching outputs. Upstream source files, public-weight revisions, checkpoint hashes, acquisition instructions, and recorded training metadata document model construction. The release adds the complete proposal grid, five H200 checkpoints, exact component-mask replay, quota and spatial-group analyses, and verification hashes.

Table 27: Computational provenance and audit controls.

Element	Verification	Archived record
Metric layers	Separate spatial, role-qualified, and queued recall	Definitions, thresholds, and integer counts
Split/merge	Sensitivity at 0.10, 0.25, and 0.50 instance overlap	Threshold grid and per-setting summaries
Scene denominator	Crop-exposure frequencies use all 26 test tiles	Tile-level queue and overlap records
Spatial grouping	Released capture indices define the spatial proxy	Date, patch, tile, and capture-index summaries
Candidate support	Rankers use the complete commissioned pool	Candidate predictions and support indicators
DINOv2 context	Paired unmasked and component-masked crops	Embedding manifests and paired queue outputs
Fine-tuning	Ten paired frozen versus last-block repeats	Seeds, checkpoints, logs, and paired differences
Target semantic model	Archived split, selection rule, and hyperparameters	Successful and unsuccessful execution records
Target semantic allocation	Fixed, complete quota-grid, date-balanced, and validation-selected policies	Per-scene selections, exact component-mask intersections, and hashes
H200 seed stability	Five independent initializations with identical data roles and optimization recipe	Checkpoints, hardware attestations, per-seed summaries, exact-mask policy replay, paired differences, and SHA-256 manifest
Raster accounting	Released-mask and paper-reported totals	Consistency and exclusion audits